STUDENT CURRICULUM

Maintenance of IO-Link Systems: Smart Sensors

CP3723 · IM4330 Training Kit

Field	Value
Course Code	CP3723
Level	Level 3
Total Time	8–10 hours
Worksheets	8
Assessment	WS8 — Final Practical Assessment
Issue	1.0
Supplier	Matrix Technology Solutions Ltd



Course Overview

What You Will Learn

- Identify and describe the function of each device in an IO-Link sensor system
- Explain the difference between basic (on/off), analogue, and IO-Link smart sensors
- Read and interpret live process data from proximity, capacitive, temperature, and light stack devices
- Use the IO-Link Master HMI dashboard to monitor port status and sensor values in real time
- Perform acyclic service data (ISDU) reads and writes to inspect and change sensor parameters
- Diagnose and correct common sensor faults: inverted output logic (NC), calibration drift, and incorrect PDout state
- Describe the role of each component in the system — sensor, master, edge device, and dashboard
- Decode basic IO-Link hex values to engineering units

System Overview

Data flows left to right: sensors send process data to the master every 500 ms, the master publishes it over MQTT to the Raspberry Pi, the Pi decodes it and pushes it to this browser via WebSocket.

Component	Model	IP Address / Connection	Role
Sensors (Ports 1–4)	Various (see below)	M12 3-wire to AL1350	Measure and report process data
IO-Link Master	IFM AL1350	192.168.7.4	Hub — all sensors plug in here
Edge Device	Raspberry Pi 4	192.168.7.2	Runs backend, serves dashboard
HMI Screen	1024×600 Touchscreen	HDMI from Pi	Live dashboard display

Equipment on the Bench

Port	Device	Model	What It Does
1	Proximity Sensor	Omron E2E-X16MB1T12	Detects metal objects — no contact, no beam
2	Capacitive Sensor	RS Pro / Carlo Gavazzi M18	Detects liquid, powder, plastic through a wall
3	Temperature Sensor	IFM TV7105	Reads temperature in °C — live value + alarms
4	Light Stack	Banner CL50 Pro	Colour/animation tower — PDout controlled
—	IO-Link Master	IFM AL1350	Hub — all sensors plug in here
—	Edge Device	Raspberry Pi 4	Runs the backend and serves this dashboard

Power Supply

- IO-Link Master and all sensors: 24V DC SELV (safety extra-low voltage — safe to touch)
- The Raspberry Pi runs at 5V via an internal DC-to-DC converter from the 24V rail
- Do not connect anything to a mains socket other than the supplied power adapter (24V).

WS1 — Meet Your Kit

Estimated Time	Dashboard	Focus
45–60 minutes	All Ports	Equipment identification, live observation, challenge

Why This Matters

Before you can maintain a system, you must be able to name every part, find it on the bench, and confirm it is working. This worksheet takes you from zero to confident — you will identify all five devices, see their live data on screen, and complete a hands-on challenge.

Learning Objectives

- Identify all components in the IM4330 kit by name, location, and function
- Confirm each sensor is live on the dashboard
- Observe the difference between on/off switching and continuous analogue data
- Navigate the HMI dashboard to view port status

Task 1 — Kit Identification Checklist

Find each item on the bench. Tick it off when you can see it and match it to the screen.

- **IO-Link Master (IFM AL1350) —** the orange box with numbered ports along the side. This is the hub. Everything else plugs into it.
- **Port 1 — Proximity Sensor (Omron E2E-X16MB1T12) —** the M18 barrel with a flat sensing face. Creates an electromagnetic field; detects metal objects with no contact and no light beam.
- **Port 2 — Capacitive Sensor —** the larger white M18 cylinder. Detects materials like liquid, powder, or plastic — even through a container wall.
- **Port 3 — Temperature Sensor (IFM TV7105) —** reads the temperature and sends back a live number in °C.
- **Port 4 — Light Stack (Banner CL50 Pro) —** the tall tower with coloured lights. Shows machine status at a glance from across the factory.
- **Edge Device (Raspberry Pi 4) —** small green circuit board in a case. Collects sensor data, runs the communications, and serves this dashboard.
- **This screen —** the live HMI dashboard. Everything the sensors say ends up here.

Task 2 — Live Check

Open the Dashboard page. Confirm all four sensor ports show a green "connected" badge. The temperature sensor should display a value close to room temperature (~20–25 °C). The proximity sensor OUT1 should be 0 (no metal in front of it).

> ℹ **NOTE** If any port shows offline — check the M12 cable connector is locked (turn locking ring clockwise until it clicks) and tell your trainer.

Task 3 — The Challenge

TASK Goal: trigger the proximity sensor (Port 1) with a metal object — but do not trigger the capacitive sensor (Port 2).

- The proximity sensor must show OUT1 = 1 (DETECTED) for at least one second
- The capacitive sensor raw value must stay below the tolerance threshold (default: 30)
- If the capacitive value exceeds the threshold, the challenge fails — click Reset and try again
- You can adjust the tolerance slider (0–10,000) to make the challenge easier or harder

NOTE What to notice: the capacitive sensor shows a live rising value as your hand gets close — this is continuous process data. A basic on/off sensor would show nothing until the threshold was crossed.

Knowledge Check

Q1. What colour is the IO-Link Master, and what is it sometimes called?

A. Orange — it is called the hub

B. Blue — it is called the controller

C. Green — it is called the gateway

Q2. Which sensor on this kit detects metal objects using an electromagnetic field?

A. The capacitive sensor on Port 2

B. The temperature sensor on Port 3

C. The proximity sensor on Port 1

Q3. What can the capacitive sensor detect that the proximity sensor cannot?

A. Objects moving faster than 1 m/s

B. Materials like liquid or powder — even through a container wall

C. The exact temperature of an object

Q4. What is the Raspberry Pi's job in this system?

A. It collects the sensor data, runs the comms, and serves up this dashboard

B. It powers the IO-Link Master via USB

C. It directly controls the light stack colours

Q5. During the challenge, what did you notice about the capacitive sensor that a normal on/off sensor wouldn't show you?

A. It only ever shows two states — on or off — same as a normal sensor

B. It measured the exact distance between your hand and the sensor in millimetres

C. It showed a live level rising as your hand got close, before it fully triggered

WS2 — What is a Smart Sensor?

Estimated Time	Dashboard	Focus
45–60 minutes	All Ports	Signal types, IO-Link animation, fault diagnosis scenario

Why This Matters

Most sensors on a factory floor are simple on/off switches. Smart sensors send back far more — a live measured value, a fault code when something goes wrong, and a device identity that the system can read automatically. This worksheet explains the three sensor types and shows you what IO-Link adds on top of a standard digital output.

Learning Objectives

- Distinguish between digital (on/off), analogue (4–20 mA), and IO-Link smart sensors
- Explain what IO-Link adds to a standard digital switching output
- Read and interpret sensor process data and fault flags on the live dashboard
- Understand the basics of MQTT and WebSocket data delivery

Background — Three Types of Sensor Signal

Type	Signal	What You Get	What You Lose
Digital (on/off)	24V / 0V switching	Simple and cheap	No value, no fault info
Analogue	4–20 mA current loop	Continuous level reading	Slow, no identity, no faults
IO-Link	Serial digital, 3-wire	Process data + faults + identity + remote config	Requires IO-Link master

ℹ **NOTE** 4–20 mA: a transmitter at 4 mA means 0% of range, at 20 mA means 100%. The midpoint (12 mA) is always 50%. This lets the receiver detect a broken cable (0 mA = below the live range).

What IO-Link adds on top of a switching output:

- The standard OUT1 switching output still works exactly as normal
- The IO-Link channel carries: live process value, fault codes, device identity (vendor, model, serial), and allows remote parameter reads and writes
- All over the same 3-wire M12 cable — no extra wires

Task 1 — Signal Animations

Open WS2 and observe the three signal type animations:

- **Digital:** Watch the LED indicator and square wave. The signal is either 0 or 24V — nothing in between.
- **Analogue:** Watch the oscilloscope. The mA value rises and falls smoothly — a continuous level.
- **IO-Link:** Watch the data packets (OUT1, Temp, Fault, Model, SP1, S/N) travel from sensor to master. Notice the square wave at the bottom — that is OUT1, still present, unchanged.

✓ **KEY POINT** Key observation: in the IO-Link animation the square wave (OUT1) continues while data packets travel above it. IO-Link does not replace the switching output — it runs alongside it.

Task 2 — Live Sensor Observation

From the Dashboard page, observe the four sensors in real time. For each sensor, note:

Sensor	What the live value shows	Process data or on/off?
Proximity (Port 1)		
Capacitive (Port 2)		
Temperature (Port 3)		
Light Stack (Port 4)		

Task 3 — Fault Diagnosis Scenario

> ⚠ **SCENARIO** Click "Start Scenario". A fault is injected into the live system — one of five possible faults. Work through the debug console clues to identify it.

The five possible faults:

1. Temperature sensor reading exceeds SP1 setpoint (Port 3)
2. Proximity sensor OUT1 stuck HIGH — on with no metal present (Port 1)
3. Wrong device identity reported on Port 2 — temperature sensor instead of capacitive
4. IO-Link communication loss on Port 3
5. Master supply voltage drops below threshold (undervoltage)

How to find the answer:

- Watch the debug console — realistic status messages stream every 300 ms
- Look for amber [WARN] lines — these appear every 8–10 entries and contain clues
- Use the clues to select the correct fault from the options provided

> ℹ **NOTE** After selecting the correct answer the scenario ends. The system restores itself automatically.

Knowledge Check

Q1. A basic digital (on/off) proximity sensor fails and its output goes silent. What diagnostic information can a technician retrieve from it?

A. Nothing — the sensor just goes silent, you have to go and test it manually

B. The exact fault code and which internal component has failed

C. The last reading before it failed, stored in its internal log

Q2. A pressure transmitter outputs 12 mA on a 4–20 mA loop calibrated 0–100 bar. What is the pressure?

A. 12 bar — the mA value equals the bar reading directly

B. 0 bar — the sensor is in fault because 12 mA is below the normal 16 mA minimum

C. 50 bar — 12 mA is the midpoint of the 4–20 mA range, which maps to 50% of 100 bar

Q3. An IO-Link sensor still has OUT1 active. What does the IO-Link channel add on top?

A. Process data, fault codes, device identity, and remote parameter writes — all over the same 3-wire cable without interrupting OUT1

B. It replaces OUT1 entirely — you only get the data stream, not a switching output

C. A higher voltage signal so the PLC can tell it is an IO-Link device

Q4. Why choose an analogue sensor over a digital sensor for tank level monitoring?

A. Analogue sensors are cheaper and easier to wire than digital sensors

B. Analogue sensors send fault codes that digital sensors cannot

C. An analogue signal gives a continuous level reading — you see exactly how full the tank is, not just "full" or "empty"

Q5. In the IO-Link wire animation, what does the square wave at the bottom of the cable represent?

A. The IO-Link data packets being encoded onto the wire

B. The standard switching output (OUT1) — still present and working exactly as a normal digital sensor, while the data stream runs above it

C. The power supply waveform showing 24V DC to the sensor

WS3 — The Proximity Sensor

Estimated Time	Dashboard	Sensor
60–75 minutes	Port 1	Omron E2E-X16MB1T12

Why This Matters

The proximity sensor is the most common sensor type in a factory. When it misbehaves — staying ON when there is nothing present, or flickering at the edge of its range — you need to be able to diagnose why without replacing it. This worksheet shows you what the sensor tells you via IO-Link, and uses ISDU to diagnose and fix a real fault.

Learning Objectives

- Read live OUT1, monitor output, instability alarm, and over-approach alarm from the dashboard
- Understand what the monitor output (amplitude) represents and what it means for sensing reliability
- Identify the Instability Alarm and Over-Approach Alarm conditions and their correct responses
- Use ISDU read to inspect the output logic parameter (index 61 / sub 1)
- Use ISDU write to correct an inverted NC output logic fault

Background — Process Data from the Omron E2E

Field	Meaning	Normal Value (free air)
OUT1	Object detected	0 (no object)
Monitor output	Oscillation amplitude — strength of the field	~80 (uint8)
Instability Alarm	Target near the edge of sensing range	0 (clear)
Over-Approach Alarm	Target too close — sensor face contact risk	0 (clear)

Output Logic — NO vs NC:

Parameter	ISDU Value	Behaviour
NO — Normal Open	0	OUT1 = 1 when object detected (default, correct)
NC — Normal Closed	1	OUT1 = 1 when NO object detected (inverted)

NOTE Output logic is stored as ISDU parameter: index 61, subindex 1, dtype uint8.

Task 1 — Live Observation

Open WS3 and observe the proximity sensor live panel. Use a metal object (screwdriver or spanner):

1. Note the OUT1 state changes — what happens when metal enters the field?
2. Bring the metal very close to the sensor face — does the Instability Alarm or Over-Approach Alarm trigger?
3. What does the monitor output value do as the metal gets closer?
4. Remove the metal — does OUT1 return to 0 immediately?

Knowledge Check (Part A)

Q1. What does the OUT1 chart do when metal is detected?

- **A.** The line jumps from 0 to 1, then drops back to 0 when the metal moves away
- **B.** The line stays flat at 0 the whole time
- **C.** The line drops below zero when metal is detected

Q2. The Instability Alarm fires while OUT1 is ON. What does this tell you?

- **A.** The sensor has detected two metal objects at the same time
- **B.** The sensor cable has a loose connection
- **C.** The target is near the edge of the sensing range — detection may be unreliable

Q3. The Over-Approach Alarm fires. What is the correct response?

- **A.** Increase the standoff distance between the sensor face and the target
- **B.** Replace the sensor immediately — the over-approach alarm means it has failed
- **C.** Reduce sensitivity using the teach button

Task 2 — NC Logic Maintenance Scenario

⚠ **SCENARIO** Work order: "Proximity sensor on Port 1 is detecting objects where none are present." Click Start Scenario to inject the fault.

Work through each step:

Step	Your Action	Expected Result
1 — Observe	Watch the OUT1 indicator. Click "I have observed the anomaly"	OUT1 shows ON with nothing present
2 — Diagnose	Select the correct fault from the three options	"Output logic changed from NO to NC"
3 — Read	ISDU Read: index 61 / sub 1	Returns value 1 (NC)
4 — Fix	ISDU Write: index 61 / sub 1, value 0 (NO)	Verify step appears
5 — Verify	Hold metal target near sensor for 3 seconds	OUT1 ON with metal, OFF without — pass
6 — Sign off	Tick all three checkboxes	Close button enables

ℹ **NOTE** Auto-restore: navigating away at any point before completing the fix will automatically write value 0 back to index 61/sub1 and restore normal operation.

WS4 — The Capacitive Sensor

Estimated Time	Dashboard	Sensor
45–60 minutes	Port 2	RS Pro / Carlo Gavazzi M18 Capacitive

Why This Matters

Capacitive sensors appear on filling lines, bin detectors, and anywhere a machine needs to sense a non-metallic material without contact. They are more sensitive than proximity sensors and easier to misadjust — a sensitivity setting that is too high will detect the container wall rather than the liquid inside. This worksheet shows you how to use IO-Link to check and correct sensitivity without physically adjusting the sensor.

Learning Objectives

- Read live detection state and raw process value from the capacitive sensor
- Understand how SP1 (Switch Point 1) controls the detection threshold
- Use the detection counter to count discrete events
- Read and write SP1 via ISDU to set the correct detection threshold
- Issue a Teach command via ISDU to set SP1 from a reference target

Background — Key ISDU Parameters

Parameter	ISDU Index	Subindex	dtype	What It Does
SP1 (Switch Point 1)	60	1	int16	Detection threshold — lower = more sensitive
QoT (Quality of Teach)	75	0	uint8	Confidence of last teach operation (0–100%)
QoR (Quality of Reference)	76	0	uint8	Confidence of reference measurement
Teach SP1	Command	—	—	Sets SP1 from current target

Task 1 — Live Detection and Counter

1. Note the raw process value in free air (nothing near the sensor face)
2. Bring your hand or a container of water near the sensor face — watch the raw value change
3. Cross the SP1 threshold — OUT1 turns ON, the detection counter increments by 1
4. Remove your hand — OUT1 returns to 0, counter holds its value

> **NOTE** A count is triggered on the OFF→ON transition (rising edge). If you hold your hand in the detection zone, you get one count regardless of how long you hold it.

Task 2 — ISDU: Read and Adjust SP1

1. Read SP1: index 60, subindex 1, dtype int16. Note the current value.
2. Compare it to the raw free-air value — is there a clear gap?
3. Write a lower value to SP1 to increase sensitivity, then a higher value to reduce it.

TASK Practical scenario: a container with powder inside is not triggering detection reliably. Lower SP1 to increase sensitivity — but not so low that the sensor detects the empty container wall.

Knowledge Check

Q1. What triggers a new count?

A. Every second while your hand is touching

B. Only when you pull your hand away

C. The moment detection goes from off to on — one touch = one count, no matter how long you hold it

Q2. Which materials can a capacitive sensor detect that a proximity sensor cannot?

A. Water and powder — even through a container wall

B. Sound and light

C. Heat and pressure

Q3. The capacitive sensor output is always ON even when nothing is near it. Most likely cause?

A. Sensitivity too high — it is detecting the container wall or nearby objects

B. The sensor needs replacing immediately

C. The cable is the wrong colour

WS5 — The Temperature Sensor

Estimated Time	Dashboard	Sensor
75–90 minutes	Port 3	IFM TV7105

Why This Matters

Temperature is the most common process measurement in manufacturing. An IO-Link temperature sensor does more than a simple thermostat — it sends back a live °C value, supports two independent alarm outputs, and stores a calibration offset that can correct sensor drift in the field without replacement.

Learning Objectives

- Read live temperature in °C and observe it as a trend
- Understand the TV7105 setpoint and reset point system (SP1, RP1, SP2, RP2)
- Adjust SP1 via slider and observe the alarm state change in real time
- Read and write SP1, RP1, SP2, RP2 via ISDU
- Diagnose and correct a calibration offset fault via ISDU write to index 681

Background — Setpoints, Reset Points, and Hysteresis

Index	Parameter	dtype	Scale	Role
583	SP1	int16	× 0.1	OUT1 switches ON when temperature rises above SP1
584	RP1	int16	× 0.1	OUT1 switches OFF when temperature falls below RP1
593	SP2	int16	× 0.1	OUT2 switches ON when temperature rises above SP2
594	RP2	int16	× 0.1	OUT2 switches OFF when temperature falls below RP2
681	Calibration Offset	int16	× 0.1	Adds/subtracts a fixed offset to the displayed temperature (±10 °C)

NOTE Hysteresis: SP must always be above its own RP. If SP1 and RP1 were the same value, the output would chatter when temperature hovers near the setpoint. Always write RP before SP — writing SP below the current RP is rejected by the sensor silently.

Task 1 — Live Temperature and Alarms

Open WS5. Observe the live temperature panel. The worksheet auto-loads dynamic setpoints based on your live temperature:

Setpoint	Formula	Example at 27.3 °C
SP2 (OUT2 pre-warning)	ceil(ambient) + 1 °C	29 °C
SP1 (OUT1 main alarm)	ceil(ambient) + 3 °C	31 °C
RP1 (OUT1 reset)	ceil(ambient)	28 °C
RP2 (OUT2 reset)	exact live temperature	27.3 °C

Hold the sensor in your hand to warm it up — observe:

1. Does OUT2 alarm activate first (at SP2)?
2. Does OUT1 alarm activate next (at SP1)?
3. When you release the sensor, does OUT1 wait until temperature drops below RP1 before resetting?

Task 2 — ISDU: Read and Write Setpoints

1. Read SP1 (index 583, sub 0, int16, scale 0.1) — note the current value
2. Read RP1 (index 584, sub 0, int16, scale 0.1)
3. Adjust SP1 to 5 °C above current temperature — confirm OUT1 alarm activates when you warm the sensor
4. Write a new RP1 value that provides at least 2 °C of hysteresis below SP1

Task 3 — Calibration Drift Scenario (Work Order CAL-0189)

⚠ **SCENARIO** Work order CAL-0189: "Port 3 temperature sensor reads approximately 3 °C above reference thermometer. No other system changes made in the past month." Click Start Scenario to inject the fault.

Step	Your Action	Expected Result
1 — Observe	Note the reading vs. actual ambient. Click "I have observed the drift"	Reading ~3 °C above actual ambient
2 — Diagnose	Select the correct fix from three options	"Write 0.0 °C — restore factory default"
3 — Read	ISDU Read: index 681 / sub 0	Returns value 3.0 (+3.0 °C offset)
4 — Fix	ISDU Write: index 681 / sub 0, value 0.0	Verify step appears
5 — Verify	Reading stays within ±1.5 °C of pre-inject baseline for 3 s	Pass
6 — Sign off	Tick all three checkboxes	Close button enables

ℹ **NOTE** Auto-restore: navigating away at any point before completing the fix will automatically write 0.0 back to index 681/sub0.

Knowledge Check

Q1. What extra can you do with an IO-Link temperature sensor compared to a basic switch?

A. Nothing extra — it works exactly the same as a switch

B. It only works at high temperatures above 100 °C

C. See the actual live temperature in °C, trend it over time, and get early warning before the trip point is reached

Q2. Move SP1 just above current temperature and hold the sensor. What should happen?

A. Nothing changes — the alarm only activates from the main dashboard

B. The alarm state turns red and shows "ABOVE THRESHOLD" once the temperature crosses your slider value

C. The sensor turns itself off to prevent overheating

Q3. Temperature reading suddenly drops to −40 °C in a room-temperature lab. Most likely cause?

A. Broken or disconnected sensor — −40 °C is the bottom of the TV7105 measurement range and appears when the sensing element is open-circuit

B. The lab is very cold

C. The setpoint has been changed

WS6 — The Light Stack

Estimated Time	Dashboard	Device
60 minutes	Port 4	Banner CL50 Pro

Why This Matters

An IO-Link light stack like the Banner CL50 Pro replaces all individual colour wires with a single 3-byte command over one M12 cable — colour, animation, speed, and intensity in one write. This worksheet shows you how that command works and what happens when the controller sends the wrong one.

Learning Objectives

- Understand PDout-only operation — the CL50 receives commands, it does not send measurements back
- Decode the 3-byte (6-hex-digit) PDout control word
- Write PDout commands to change colour, animation, intensity, and speed
- Diagnose a light stack showing the wrong state after a maintenance restart

Background — PDout Control Word

Byte	Bits	Field	Values
Octet 0	[7:6]	Audible	0=Off, 1=On, 2=Pulsed, 3=SOS
Octet 0	[5:3]	Colour 2 Intensity	0=High, 1=Low, 2=Medium, 3=Off
Octet 0	[2:0]	Colour 1 Intensity	0=High, 1=Low, 2=Medium, 3=Off
Octet 1	[7:6]	Speed	0=Medium, 1=Fast, 2=Slow
Octet 1	[5:3]	Pulse Pattern	0=Normal, 1=Strobe, 2=Three Pulse, 3=SOS, 4=Random
Octet 1	[2:0]	Animation	0=Off, 1=Steady, 2=Flash, 3=Two Colour Flash, 4=Intensity Sweep
Octet 2	[7:4]	Colour 2 index	0=Green, 1=Red, 2=Orange, 3=Amber, 4=Yellow … 9=Blue … 13=White
Octet 2	[3:0]	Colour 1 index	0=Green, 1=Red, 2=Orange, 3=Amber, 4=Yellow … 9=Blue … 13=White

NOTE Default startup state: 000100 — Green, Steady, High intensity. This is written automatically when the system starts.

Task 1 — Quick Colour Buttons

1. Set the light stack to Red, Steady
2. Set it to Amber, Flash
3. Set it back to Green, Steady (the default safe state)

Task 2 — Full Control Grid (Decode Exercise)

Open the CL50 configuration modal and set: Colour 1 = Blue, Animation = Two Colour Flash, Colour 2 = White, Speed = Slow. Observe the live hex preview before clicking Save.

Decode exercise — what does 180101 mean?

- Octet 0 = 0x18 → Colour 1 Intensity bits [2:0] = 0 (High), Colour 2 bits [5:3] = 3 (Off)
- Octet 1 = 0x01 → Animation bits [2:0] = 1 (Steady), Speed = Medium
- Octet 2 = 0x01 → Colour 1 low nibble = 1 (Red)

Task 3 — Maintenance Scenario LST-0312

> ⚠ **SCENARIO** Work order LST-0312: "Light stack on production cell 4 is showing amber flash. Maintenance team has cleared the fault and confirmed the machine is running normally. Light stack will not return to green." Observe the CL50 and select the correct diagnosis.

Select the correct diagnosis:

- **A:** The CL50 hardware is malfunctioning — the amber segment is stuck and cannot be cleared by software
- **B:** IO-Link comms to Port 4 have dropped — the light is locked to its last received command and cannot be updated
- **C:** The PLC output register is still holding the pre-maintenance fault PDout value (184203 — Amber Flash). The normal-run value (000100 — Green Steady) was never sent after the PLC restarted

Knowledge Check

Q1. The CL50 is "PDout-only". What does that mean?

A. The master sends a command value to the light — no measurement data comes back from the device to the master

B. It only works with certain IO-Link masters

C. It uses a separate digital output wire for each colour segment

Q2. The hex value 180101 is sent to the CL50. Which colour will it show?

A. Green — Octet 2 low nibble = 0x00

B. Amber — 0x18 in Octet 0 indicates Amber

C. Red — Octet 2 low nibble = 0x01 = Red (index 1)

Q3. Flashing red will not turn off after a fault is cleared. Most likely cause?

A. The PLC is still sending a "red on" PDout command — the fault state is held in the controller, not the light itself

B. The light stack hardware is broken

C. IO-Link has lost connection to the light stack

Q4. Why is IO-Link PDout better than a traditional 3-wire digital light stack?

A. It is always cheaper per unit due to IO-Link certification reducing manufacturing costs

B. The colour cannot be changed once wired — it is safer

C. Full control of colour, animation, intensity, and speed over a single 3-pin cable — plus the HMI can read back the exact state without a separate feedback wire

WS7 — Fault Finding and Replacement

Estimated Time	Dashboard	Focus
60–75 minutes	All Ports	Semi-independent fault diagnosis — no answers given

Why This Matters

This worksheet is the closest thing to a real fault-finding call-out. A fault has been reported — the light stack gives you the first clue before you check anything else. You work through the system logically: check the light stack state, check the IO-Link Master page, identify the affected port, diagnose via ISDU, and write the fix. No answer is given — you must work it out.

Learning Objectives

- Use the light stack as a first diagnostic indicator before examining individual sensors
- Navigate the IO-Link Master page to identify abnormal port behaviour
- Apply ISDU read and write to diagnose and correct an output logic fault
- Follow a systematic fault-finding process: observe → isolate → diagnose → fix → verify

Recall

ℹ **NOTE** IO-Link sensors use NO (Normal Open) or NC (Normal Closed) output logic, stored at ISDU index 61 / subindex 1.
NO (value 0): OUT1 = 1 when an object is detected (default, correct behaviour).
NC (value 1): OUT1 = 1 when NO object is detected (inverted — looks like a permanent fault).

The Scenario

⚠ **SCENARIO** Work order: "Production cell: object presence sensor on conveyor is showing an object present at all times, even when the conveyor is empty." No other information is given.

Step	Your Action	Guidance
1 — Check the light stack first	Open the dashboard — look at Port 4	What state is the CL50 showing? This is how a maintenance technician starts — the andon light tells you something is wrong before you check individual sensors.
2 — Open the IO-Link Master page	Check each port	Which port is showing an unexpected state? What does OUT1 show with no metal object in front of the sensor?
3 — Diagnose	Select the correct fault from three options (A / B / C)	Consider IO-Link comms drop, hardware failure, and output logic fault.
4 — Fix	Write the correct ISDU value	Index 61 / sub 1 — what value restores normal NO operation?
5 — Verify	Confirm correct behaviour	OUT1 = 0 with no metal present. OUT1 = 1 when metal is brought close. Light stack returns to normal run state.

ℹ **NOTE** There is no MCQ in this worksheet — the scenario IS the assessment. Work through it systematically without asking for the answer.

WS8 — Final Practical Assessment

Estimated Time	Format	Notes
45–60 minutes	Individual	Tasks 2A, 3, and 4 are completed without trainer support

⚠ **SCENARIO** This is your individual assessment for CP3723. Tasks 2A, 3, and 4 are unaided — no trainer support. Task 1 (device identification) may be supported.

Quick Reference

Concept	Rule / Values
ISDU hex decode	Raw hex → decimal → multiply by scale factor = engineering value Example: 01 F4 → 500 decimal → × 0.1 = 50.0 °C
Negative int16	Values above 0x7FFF (32767) are negative — use two's complement
CL50 Colour Index	0=Green, 1=Red, 2=Orange, 3=Amber, 4=Yellow, 5=Lime, 6=Spring Green, 7=Cyan, 8=Sky Blue, 9=Blue, 10=Violet, 11=Magenta, 12=Rose, 13=White

Task 1 — Device Identification

Open the IO-Link Master page. For each port, record the device name, Vendor ID, and device type shown on the dashboard.

Port	Device Name (from dashboard)	Vendor ID	Device Type
1			
2			
3			
4			

Task 2A — ISDU Hex Decode

The temperature sensor on Port 3 returns the following value when SP1 (index 583, subindex 0) is read via ISDU:

```
Raw hex value:  01 F4     dtype: int16     scale: × 0.1
```

What is the setpoint in °C? Show your working.

Working:

Answer: _____ °C

Task 3 — PDout Colour Identification

The following PDout value is written to the CL50 light stack on Port 4:

```
PDout:  000109
```

What colour will the light stack show? Show your working using the Octet 2 colour table above.

Working:

Answer: _____

Task 4 — Knowledge Check

Answer all four questions individually without support.

Q1. The Omron E2E instability alarm fires but OUT1 is still switching correctly. What is the right course of action?

A. Ignore it — the output is working so there is no problem

B. Replace the sensor immediately

C. Investigate — instability means the target is at the edge of sensing range. Small vibrations will cause intermittent output. Reposition the target or adjust the sensing distance before it causes a fault

Q2. After replacing the capacitive sensor on Port 2, its output is ON even with nothing near it. Re-teaching did not fix it. What is the most likely cause?

A. Output logic is set to NC — the sensor output is inverted so it is ON when nothing is present. Read the output logic parameter via ISDU to confirm, then write NO (0) to correct it

B. The IO-Link cable polarity is reversed — swap Pin 2 and Pin 4 at the master connector

C. SP1 is set correctly — the container wall is triggering the output and this is expected behaviour for a capacitive sensor

Q3. A temperature sensor reading drifts 3 °C high after months in service. No other system changes were made. What is the correct IO-Link field action?

A. Replace the sensor — drift always indicates a faulty unit

B. Write a −3.0 °C calibration offset to the sensor via ISDU (index 681) to correct the reading in the field, verified against a reference thermometer

C. Adjust SP1 down by 3 °C to compensate for the drift

Q4. What is the key advantage of the IO-Link light stack over a conventionally wired one?

A. IO-Link light stacks use less power

B. IO-Link light stacks can display more colours than conventional ones

C. All colours, animation patterns, and intensity levels are set by a single 3-byte PDout value over one standard cable. Conventional wiring needs a separate signal wire per segment

Further Study

The Further Study section (accessible from the sidebar) provides eight self-paced chapters with interactive games, simulations, and knowledge checks.

Chapter	Title	Key Takeaway
1	Industry 4.0 & IoT	Less guessing, more data — use sensor data to plan repairs before failure
2	Smart Sensors & IO-Link	One cable: process data, parameters, diagnostics, and identity
3	4–20 mA Analogue Signals	12 mA = 50% of range — the mA value is not the engineering value
4	IO-Link in Depth	ISDU = acyclic service data; PDin/PDout = cyclic process data
5	Sensor Types and Selection	Proximity vs. capacitive vs. optical — when to use each
6	Temperature Measurement	PT100 resistance curve, calibration drift, thermocouple vs. RTD
7	Electromagnetics	How inductive sensors generate and detect field disturbance
8	Device Identity	Vendor ID, Device ID, IODD — how the master auto-identifies a sensor

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Assessment Summary

Task	Format	Type	Notes
WS1–WS6 MCQs	Multiple choice (in-worksheet)	Formative	Immediate feedback on screen
WS3 NC Logic Scenario	Step-by-step ISDU task	Formative	Auto-restore on navigation
WS5 Calibration Scenario	Step-by-step ISDU task	Formative	Auto-restore on navigation
WS6 LST-0312 Scenario	Diagnosis selection	Formative	—
WS7 Fault Finding	Full scenario — no hints	Formative	Semi-independent
WS8 Task 1	Device identification	Summative	Trainer may support
WS8 Task 2A	Hex decode — free response	Summative	Unaided
WS8 Task 3	PDout colour — free response	Summative	Unaided
WS8 Task 4	4-question MCQ	Summative	Unaided

Your progress through all worksheets is tracked on the Student Progress page (sidebar). You can export a JSON record of your session for your portfolio.